Stochastic Process

A random variable $X \sim F_X(x)$

A stochastic process $\{X_1, X_2, X_3, ..., X_t, ...\}$

The time index t can be discrete/ continuous.

The "simplest" stochastic process is: all X_t's are independent and identically distributed

The "simplest" yet "non-trivial" stochastic process is the Markov Process

$$\mathbb{P}\{X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, ..., X_0 = i_0\} = \mathbb{P}\{X_{n+1} = j | X_n = i\}$$

for all $i_0, i_1, ..., i_{n-1}, i, j$, and all $n \geq 0$

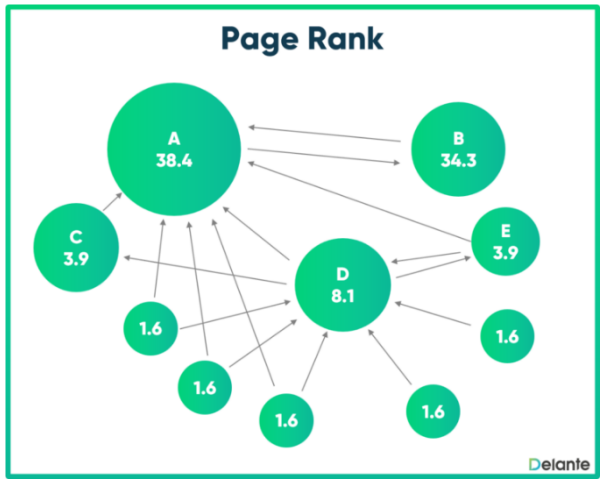
X_n: the state of the system at time n

Examples of Markov Chain

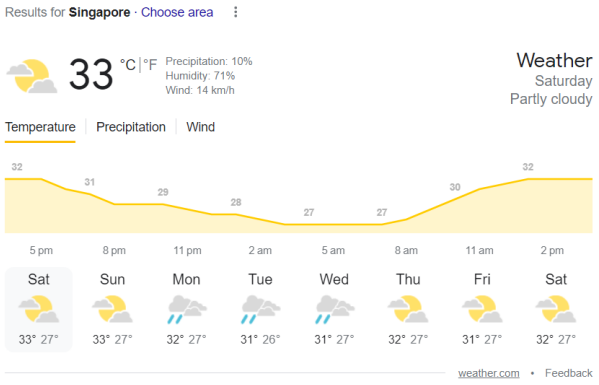
Board Games: Monopoly, Snakes and Ladders



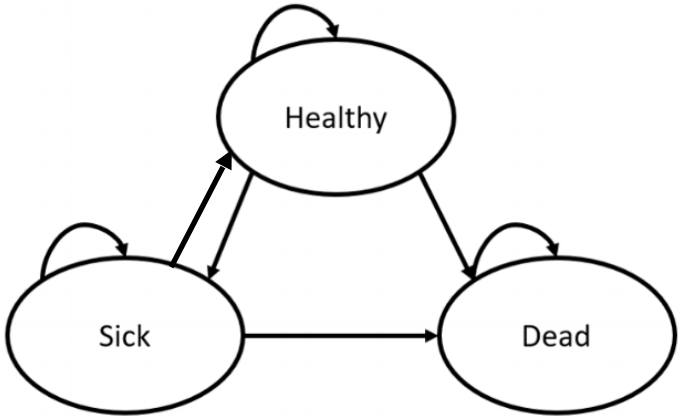
PageRank Algorithm



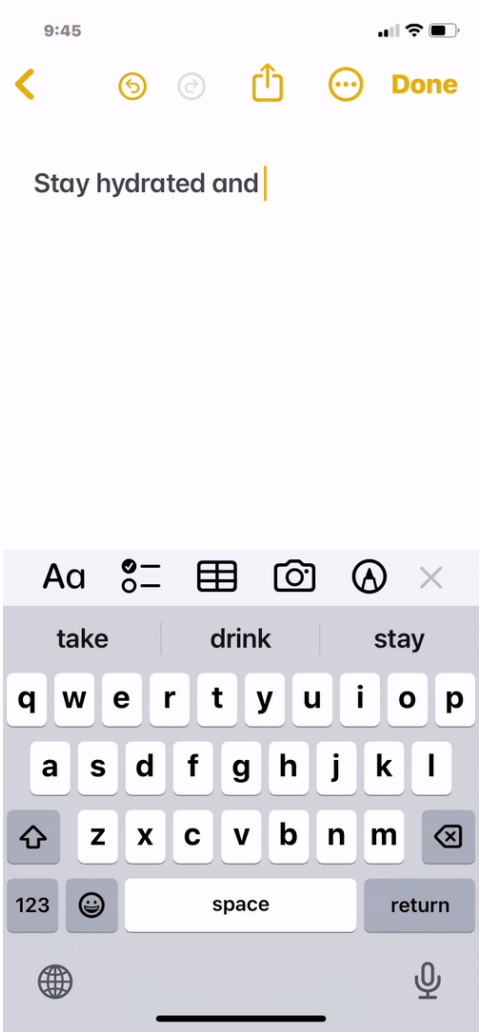
Weather Forecasting



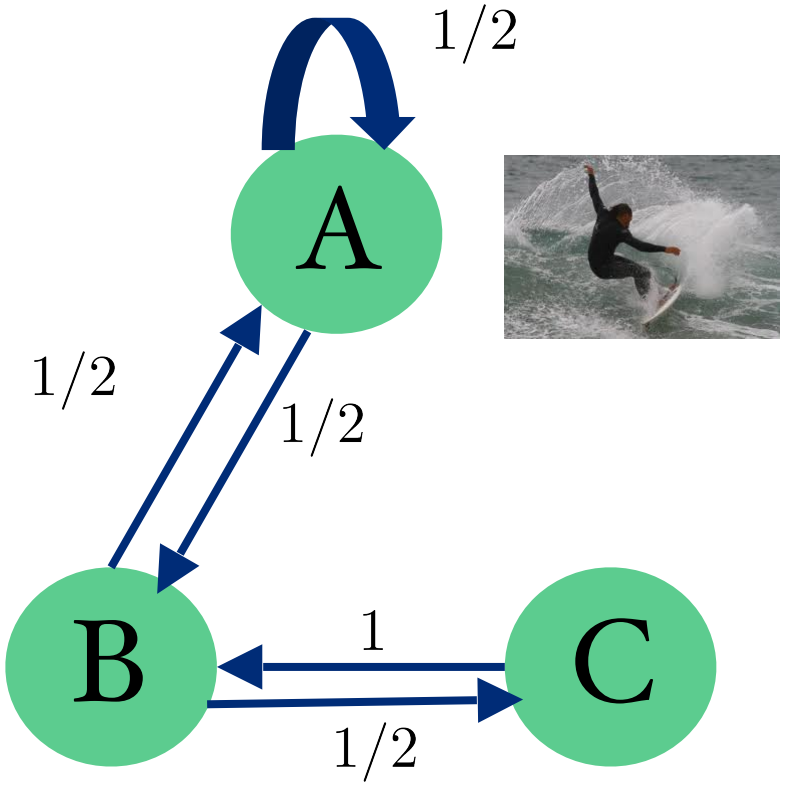
Health State Modeling



Predictive Text and Keyboard



PageRank: Random Walk Interpretation



WWW in 400 B.C.

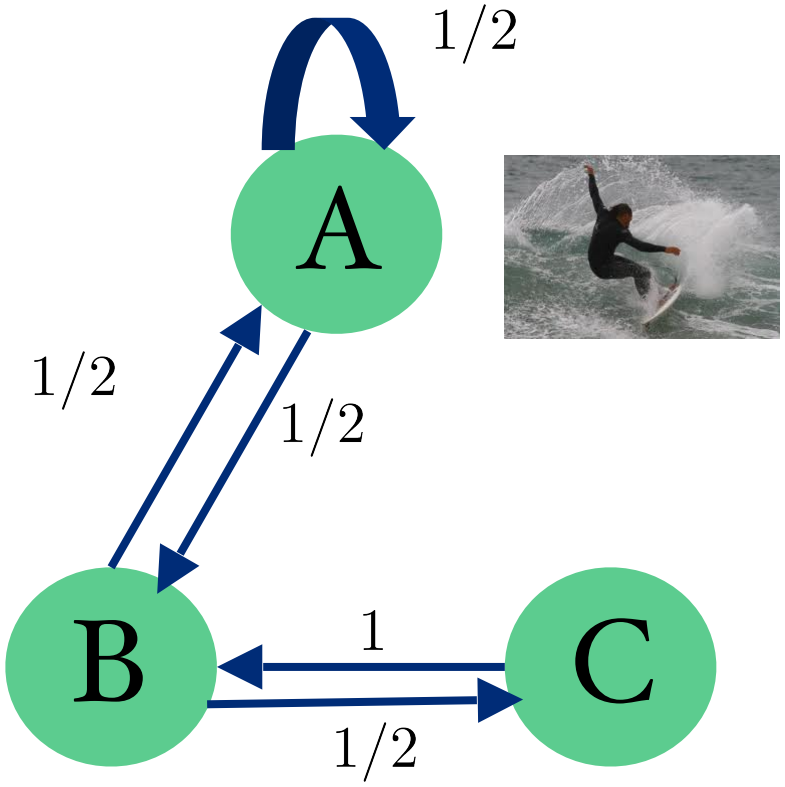
When you are reading a webpage, you see hyperlinks in the article, then you click it, you land on another webpage, where you see hyperlinks and you click…

This behavior can be modeled as a "Random Walk"

- You are on a connected graph
- At time *t*, you are on a page
- At time *t*+1, you are on one of the pages that are pointed by the page you read at time *t*, with equal probability
- This process continues indefinitely

What is the probability that I will be at page *C* after 5 steps since I started my surfing at page *A*?

PageRank: Random Walk Interpretation (Markovian Property)



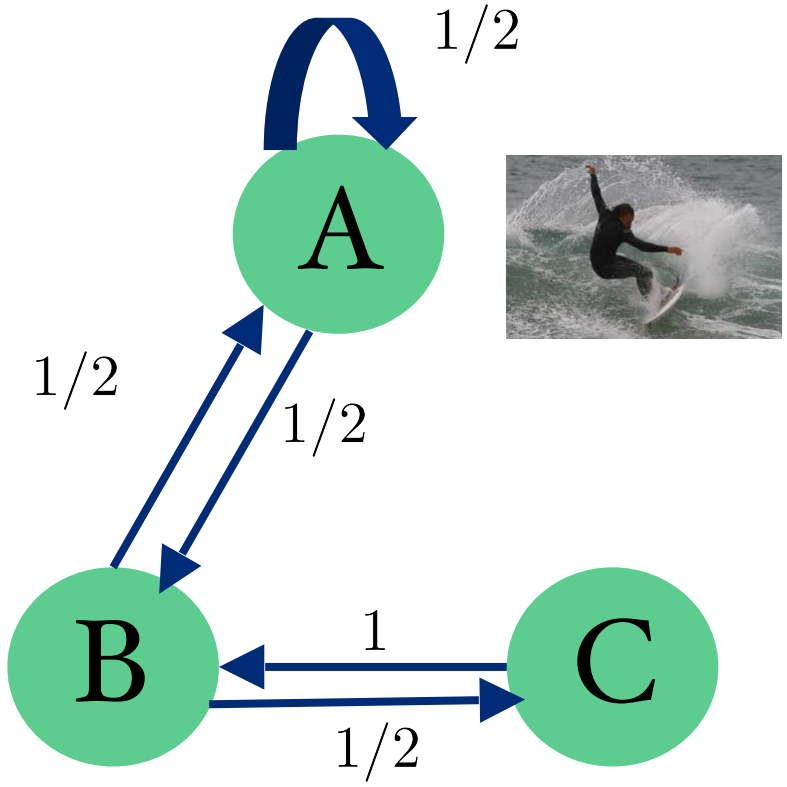
WWW in 400 B.C.

The Graph describes the real WWW, and one critical assumption is made on the user web surfing behavior

The probability that a user lands on another page after one step only depends on the current page s/he is on

Markovian Property

PageRank: Random Walk Interpretation (1-Step Transition Probability Matrix)



WWW in 400 B.C.

Imagine that initially, you have equal chance to open any one of the pages in the network

Markovian assumption on users' web surfing behavior allows us to answer the question like "What is the probability that I will be at page C after 5 steps since I started my surfing at page A?"

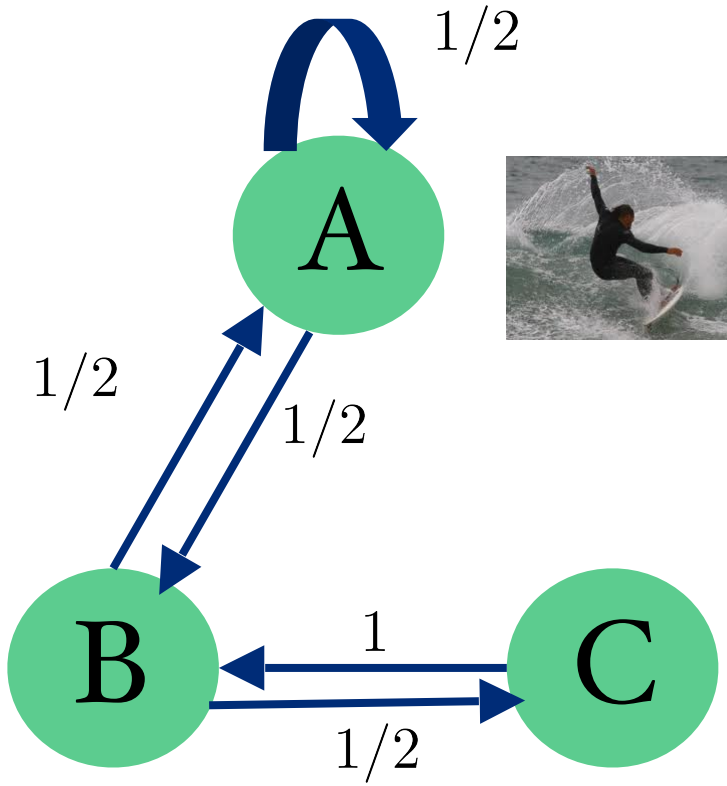
$$M = \begin{array}{c} \\ \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{array} \begin{array}{c} \begin{array}{ccc} \mathbf{A} & \mathbf{B} & \mathbf{C} \end{array} \\ \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1 \\ 0 & 1/2 & 0 \end{bmatrix} \end{array}$$

(1-step) Transition Matrix

$$\begin{bmatrix} r_A \\ r_B \\ r_C \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \quad \mathbf{p}$$

What is the probability that you will be at page A, B, and C after one step, given your initial probability distribution **p** over all websites in the network?

PageRank: Random Walk Interpretation (1-Step Transition Probability Matrix)





WWW in 400 B.C.

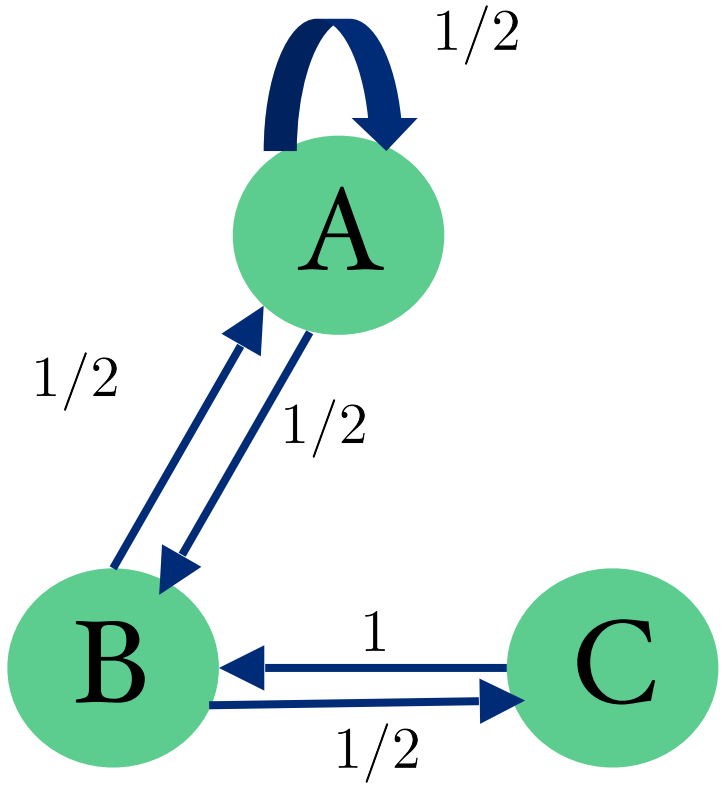
Markovian assumption on users' web surfing behavior allows us to answer the question like "What is the probability that I will be at page C after 5 steps since I started my surfing at page A?"

$$\begin{array}{c} \\ \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{array} \begin{array}{c} \begin{array}{ccc} \mathbf{A} & \mathbf{B} & \mathbf{C} \end{array} \\ \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1 \\ 0 & 1/2 & 0 \end{bmatrix} \end{array} \begin{array}{c} \mathbf{p} \\ \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \end{array} = \begin{bmatrix} 1/3 \\ 1/2 \\ 1/6 \end{bmatrix}$$

(1-step) Transition Matrix

$$\mathbb{P}\{X_{t+1} = A | X_t = A\} \cdot \mathbb{P}\{X_t = A\} + \mathbb{P}\{X_{t+1} = A | X_t = B\} \cdot \mathbb{P}\{X_t = B\} + \mathbb{P}\{X_{t+1} = A | X_t = C\} \cdot \mathbb{P}\{X_t = C\}$$

PageRank: Random Walk Interpretation (1-Step Transition Probability Matrix)



$$\begin{array}{c} \\ \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{array}\begin{array}{c} \begin{array}{ccc} \mathbf{A} & \mathbf{B} & \mathbf{C} \end{array} \\ \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1 \\ 0 & 1/2 & 0 \end{bmatrix} \end{array} \cdot \begin{array}{c} \mathbf{p} \\ \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \end{array} = \begin{bmatrix} 1/3 \\ 1/2 \\ 1/6 \end{bmatrix}$$

$$\begin{array}{cccccccc} \mathbf{p} & & \mathbb{M}\mathbf{p} & & \mathbb{M}^2\mathbf{p} & & \mathbb{M}^3\mathbf{p} & & \mathbb{M}^t\mathbf{p} & & \mathbb{M}^\infty\mathbf{p} \\ \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} & \begin{bmatrix} 1/3 \\ 1/2 \\ 1/6 \end{bmatrix} & \begin{bmatrix} 5/12 \\ 1/3 \\ 1/4 \end{bmatrix} & \begin{bmatrix} 3/8 \\ 11/24 \\ 1/6 \end{bmatrix} & \cdots & \begin{bmatrix} \dots \\ \dots \\ \dots \end{bmatrix} & \cdots & \begin{bmatrix} 2/5 \\ 2/5 \\ 1/5 \end{bmatrix} \end{array}$$

Step	**0**	**1**	**2**	**3**	**…**	***t***	**…**	∞

WWW in 400 B.C.

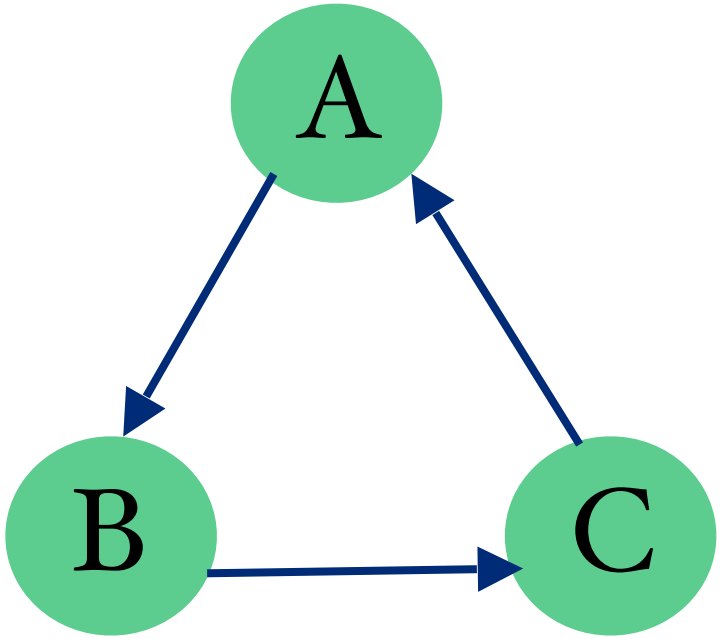
$\mathbb{M}^t\mathbf{p}$ is the probability distribution over pages after t steps transition given the initial probability distribution $\mathbf{p}$.

PageRank: Two Questions

$$\mathbf{p}, \mathbb{M}\mathbf{p}, \mathbb{M}^2\mathbf{p}, \mathbb{M}^3\mathbf{p}, \ldots, \mathbb{M}^t\mathbf{p}, \ldots$$

- Does this always converge (condition)?
- Does it converge to something meaningful?

“Periodicity” Problem



$$\begin{array}{c} \\ \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{array} \begin{array}{c} \begin{array}{ccc} \mathbf{A} & \mathbf{B} & \mathbf{C} \end{array} \\ \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \end{array}$$

$$\begin{array}{cccccc} \mathbf{p} & \mathbb{M}\mathbf{p} & \mathbb{M}^2\mathbf{p} & \mathbb{M}^3\mathbf{p} & & \\ \begin{bmatrix} r_A \\ r_B \\ r_C \end{bmatrix} & \begin{bmatrix} r_C \\ r_A \\ r_B \end{bmatrix} & \begin{bmatrix} r_B \\ r_C \\ r_A \end{bmatrix} & \begin{bmatrix} r_A \\ r_B \\ r_C \end{bmatrix} & \cdots & \\ \textbf{Iteration} & \textbf{0} & \textbf{1} & \textbf{2} & \textbf{3} & \textbf{...} \end{array}$$

In general, a state is periodic if returning to it has positive probability only at multiples of some integer (d>1). The return probability need not be 1. The largest such d is its period.

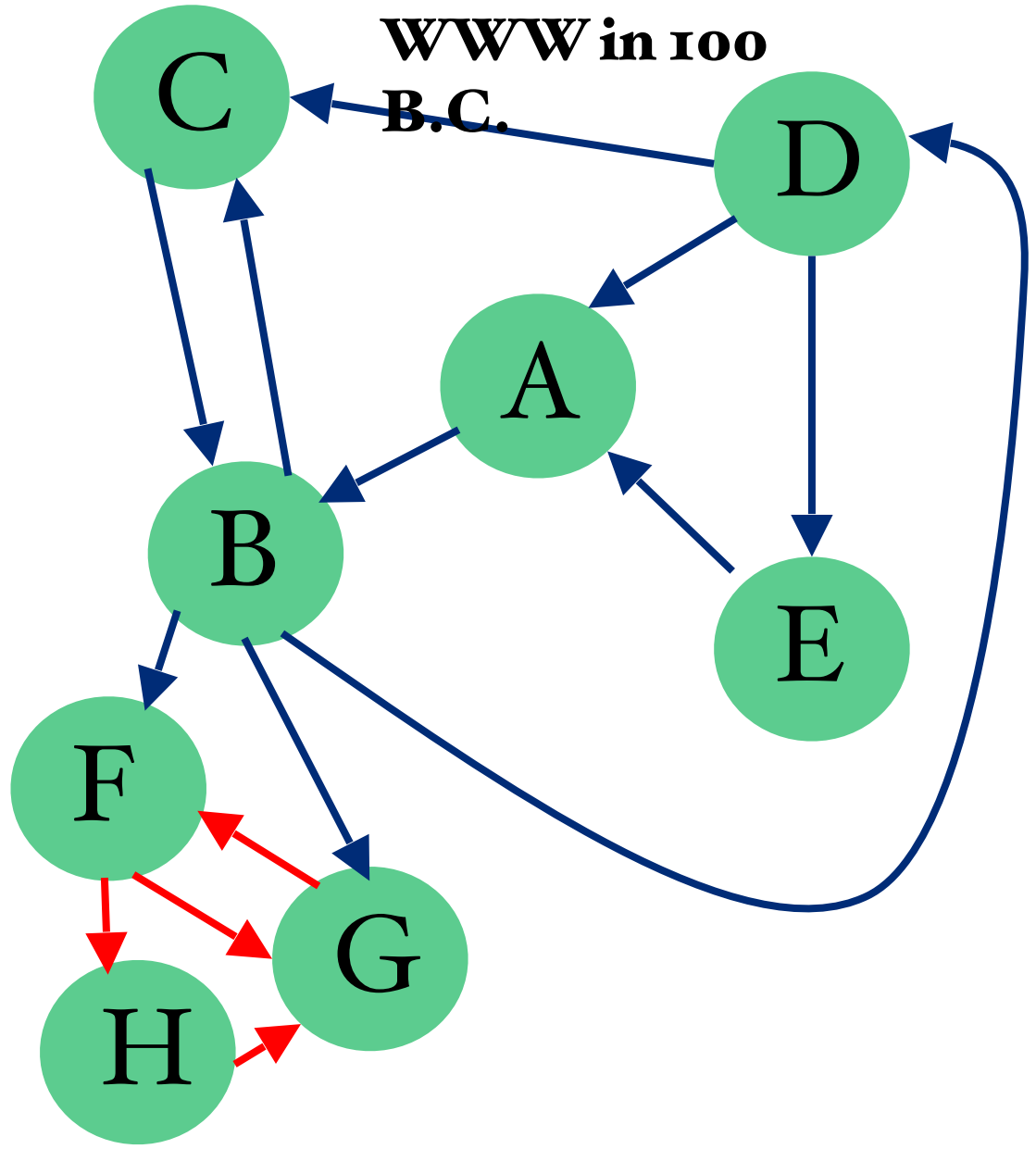
"Spider Traps" Problem

Imagine a random walk on this graph. Whenever the walk lands on F or G, it is "stuck" on the subgraph F, G, and H

	A	B	C	D	E	F	G	H
A	0	0	0	1/3	1	0	0	0
B	1	0	1	0	0	0	0	0
C	0	1/4	0	1/3	0	0	0	0
D	0	1/4	0	0	0	0	0	0
E	0	0	0	1/3	0	0	0	0
F	0	1/4	0	0	0	0	1	0
G	0	1/4	0	0	0	1/2	0	1
H	0	0	0	0	0	1/2	0	0

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 2/5 \\ 2/5 \\ 1/5 \end{bmatrix}$$

$$\mathbf{p}, \mathbb{M}\mathbf{p}, \mathbb{M}^2\mathbf{p}, \mathbb{M}^3\mathbf{p}, \ldots, \mathbb{M}^t\mathbf{p}, \ldots$$



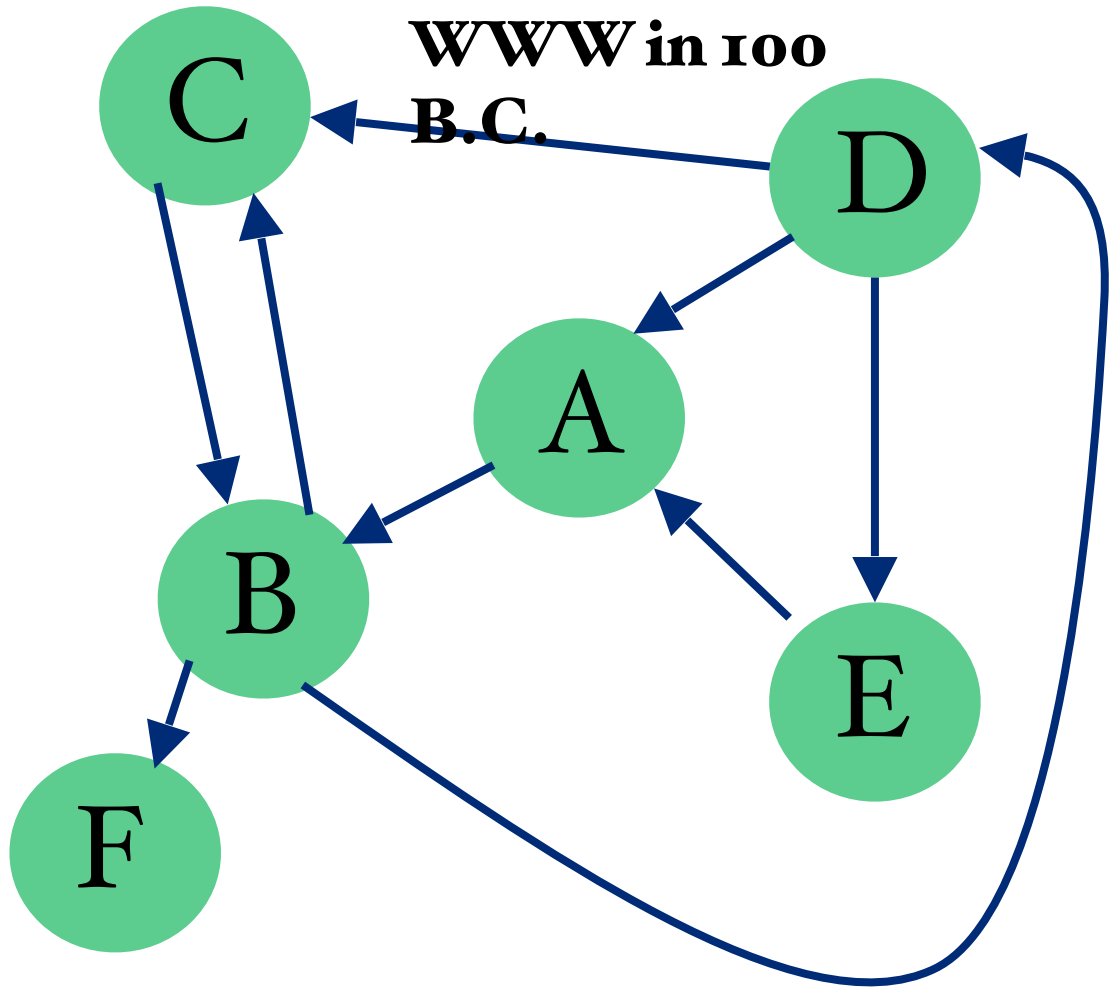
"Dead End Node" Problem

Eventually, all flows goes to page F and F has no outbound link. In particular, F does not even point to itself

$$\mathbf{p}, \mathbb{M}\mathbf{p}, \mathbb{M}^2\mathbf{p}, \mathbb{M}^3\mathbf{p}, \ldots, \mathbb{M}^t\mathbf{p}, \ldots$$

$\longrightarrow$ 0, as t goes to infinity

There are dead-end nodes in this graph, and eventually all websites in the network have a PageRank score of 0.



Larry Page's Solution: Random Teleporting

foodpanda.com

train cat

Useful tips to train your cat

October 07, 2020

First things first: Never punish

Cats simply won't learn from what some owners would consider "discipline." Worse yet, "punishing" your cat can induce stress, leading to behavioral and health problems—not something you want to deal with in cat training. Remember that patience and positive reinforcement are essential if you're learning how to train a cat. Trying to figure out your cat's behavior?



Larry Page's Solution: Random Teleporting

where to buy durian

- **where to buy durian** — Search with Google
- **where to buy durian** in singapore
- **where to buy durian** in jb
- **where to buy durian** in tampines
- **where to buy durian** in sengkang
- **where to buy durian** in hougang
- **where to buy durian** in yishun
- **where to buy durian** in woodlands
- **where to buy durian** cake in singapore
- **where to buy durian** near me

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Special pages
Permanent link
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Cite this page

In other projects

Wikimedia Commons
Wikiquote

The band's 1977 album *News of the World* contained "We Will Rock You" and "We Are the Champions", which have become anthems at sporting events. By the early 1980s, Queen were one of the biggest stadium rock bands in the world. "Another One Bites the Dust" (1980) became their best-selling single, while their 1981 compilation album *Greatest Hits* is the best-selling album in the UK and is certified eight times platinum in the US. Their performance at the 1985 Live Aid concert has been ranked among the greatest in rock history by various publications. In August 1986, Mercury gave his last performance with Queen at Knebworth, England. In 1991, he died of bronchopneumonia - a complication of AIDS, and Deacon retired in 1997. Since 2004, May and Taylor have toured under the "Queen +" name with vocalists Paul Rodgers and Adam Lambert.

Estimates of Queen's record sales range from 170 million to 300 million records, making them one of the world's best-selling music artists. In 1990, Queen received the Brit Award for Outstanding Contribution to British Music from the British Phonographic Industry. They were inducted into the Rock and Roll Hall of Fame in 2001. Each member has composed hit singles, and all four were inducted into the Songwriters Hall of Fame in 2003. In 2005, Queen received the Ivor Novello Award for Outstanding Song Collection from the British Academy of Songwriters, Composers, and Authors. In 2018, they were presented the Grammy Lifetime Achievement Award.

Contents [hide]

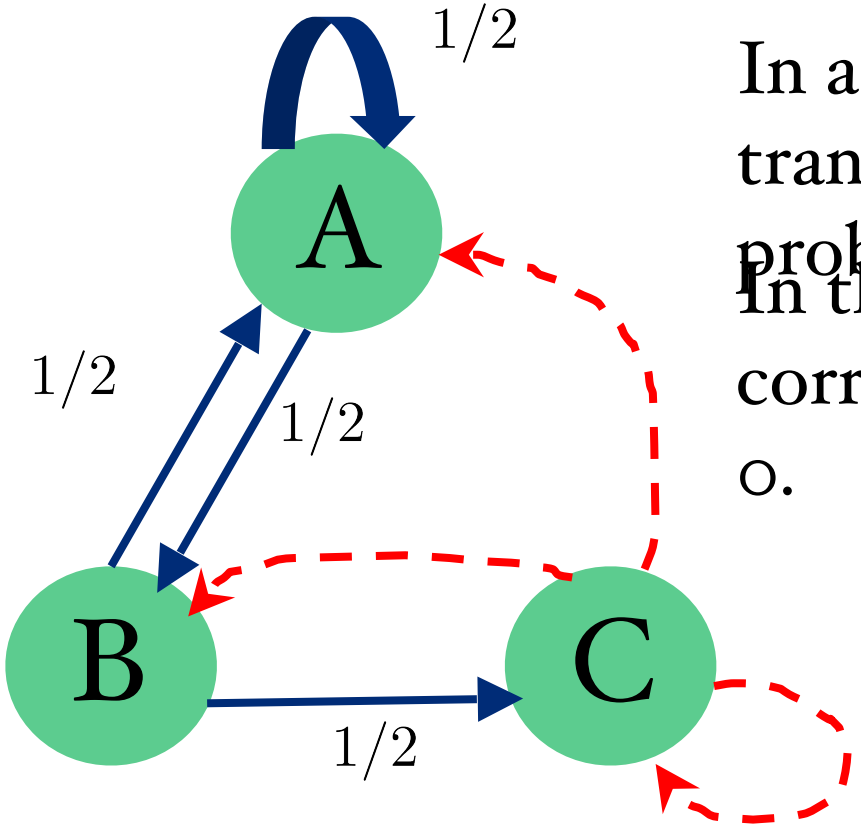
1 History
1.1 1968–1974: Early era
1.2 1974–1976: *Sheer Heart Attack* to *A Night at the Opera*



Top: Brian May, Freddie Mercury
Bottom: John Deacon, Roger Taylor

Background information	
Origin	London, England
Genres	Rock
Years active	1970–present
Labels	EMI · Parlophone · Elektra · Capitol · Hollywood · Island · Virgin EMI
Associated acts	Smile · Queen + Paul Rodgers · Queen + Adam Lambert · David Bowie · Tim Staffell
Website	queenonline.com

Modifications (Solve "Dead-end" Node Issue)



In a graph with "dead-end" nodes, the one-step transition probability matrix is not a "valid probability matrix".

In the "dead-end" case, the column in the matrix M that corresponds to the dead-end node has all entries being 0.

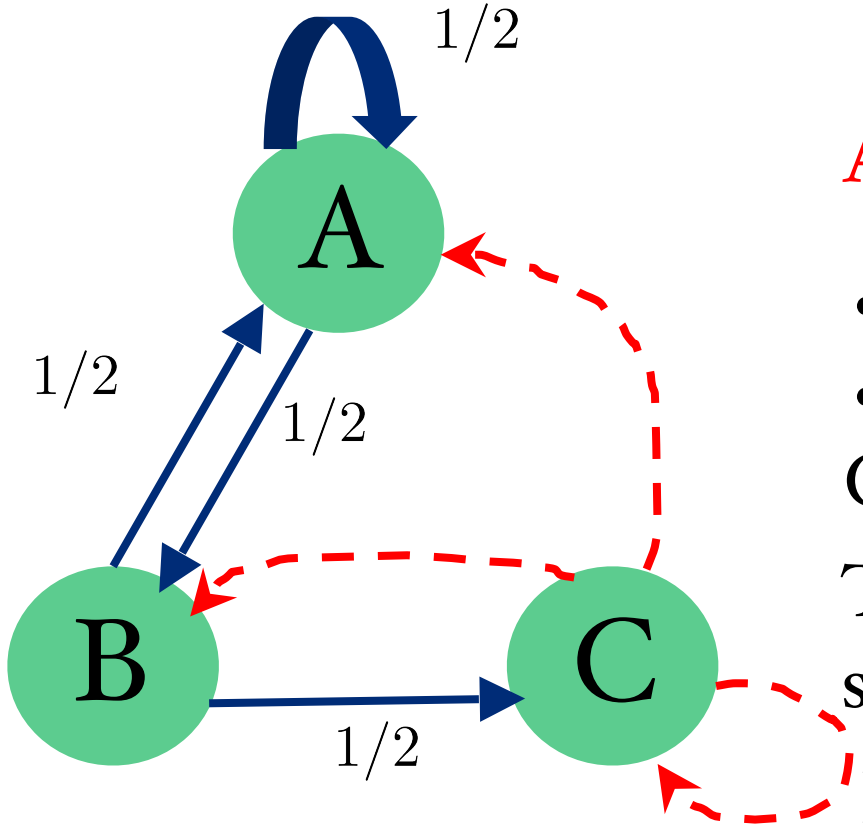
$$\mathbb{H} = \mathbb{M} + \frac{1}{N}\mathbf{e} \cdot \mathbf{a}^T \qquad \mathbf{e} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix},$$

$$a_i = \begin{cases} 1, & \text{if node } i \text{ is dead end} \\ 0, & \text{otherwise} \end{cases} \qquad \mathbf{a} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

$$\mathbf{e} \cdot \mathbf{a}^T = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

	A	**B**	**C**
A	1/2	1/2	1/3
B	1/2	0	1/3
C	0	1/2	1/3

Modifications (Solve "Spider Trap" and "Periodicity" Issues)



At each step, the random walker has 2 options:

- With probability β, follow a link at random;
- With probability $1-\beta$, jump to some random page;

Common value for $\beta \in [0.8, 0.9]$

The surfer will teleport out of spider trap within some steps

It can also be verified that there are no periodicity issues anymore after this random teleportation modification.

Summary

- **The final form of Online Surfing Model for Deriving PageRank Scores**

$$\mathbb{G} = \beta\mathbb{H} + (1-\beta)\frac{1}{n}\mathbf{e}\cdot\mathbf{e}^T \quad \text{where } \mathbb{H} = \mathbb{M} + \frac{1}{n}\mathbf{e}\cdot\mathbf{a}^T$$

$$\mathbb{G} = \beta\mathbb{M} + \frac{\beta}{n}\mathbf{e}\cdot\mathbf{a}^T + (1-\beta)\frac{1}{n}\mathbf{e}\cdot\mathbf{e}^T$$

This matrix describes a fully connected graph where any node can reach any node with equal probability

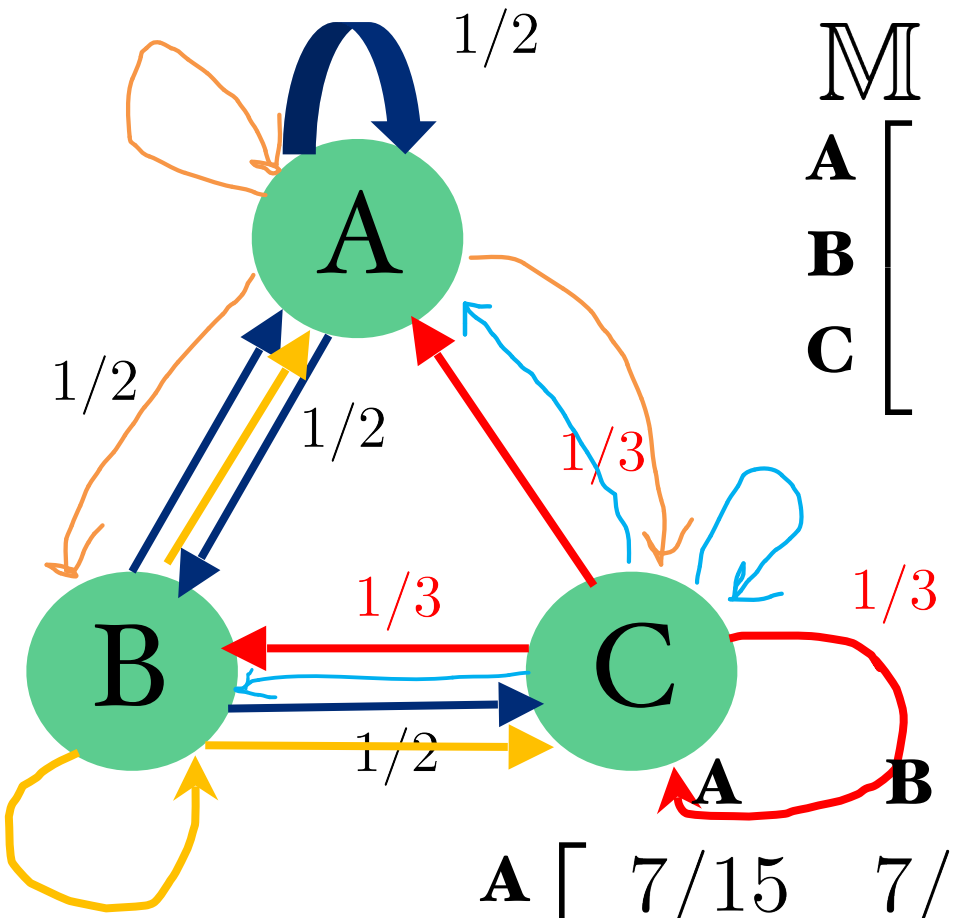
A web surfer randomly picks links to follow

A web surfer can't randomly pick a link when they land on a page with no outbound links

A web surfer gets bored and teleports to a random webpage

$\mathbf{p}, \mathbb{G}\mathbf{p}, \mathbb{G}^2\mathbf{p}, \mathbb{G}^3\mathbf{p}, \ldots, \mathbb{G}^t\mathbf{p}, \ldots$ will converge to a reasonable probability vector

Random Teleports (beta = 0.8)



$$\mathbb{M} = \begin{array}{c} \\ \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{array} \begin{array}{c} \begin{array}{ccc} \mathbf{A} & \mathbf{B} & \mathbf{C} \end{array} \\ \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 0 \\ 0 & 1/2 & 0 \end{bmatrix} \end{array}$$

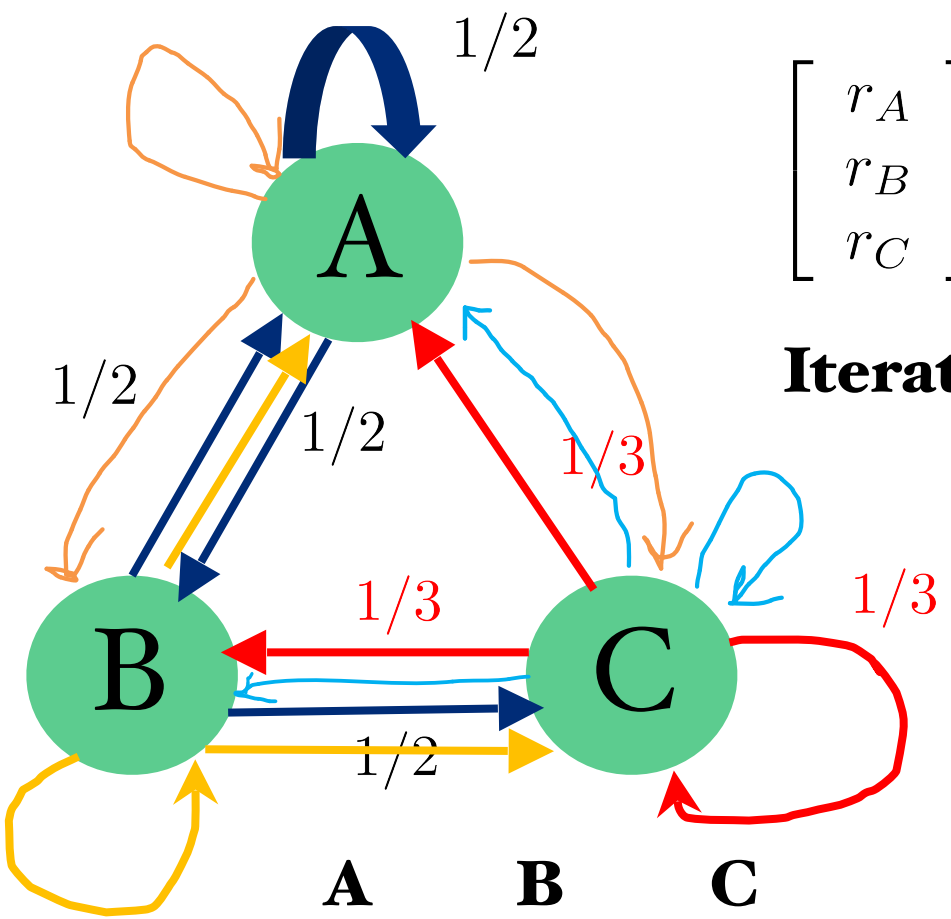
$$\mathbb{H} = \begin{array}{c} \\ \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{array} \begin{array}{c} \begin{array}{ccc} \mathbf{A} & \mathbf{B} & \mathbf{C} \end{array} \\ \begin{bmatrix} 1/2 & 1/2 & 1/3 \\ 1/2 & 0 & 1/3 \\ 0 & 1/2 & 1/3 \end{bmatrix} \end{array}$$

$$\mathbb{H} = \mathbb{M} + \frac{1}{N}\mathbf{e} \cdot \mathbf{a}^T$$

$$\mathbb{G} = \beta\mathbb{H} + (1-\beta)\frac{1}{n}\mathbf{e} \cdot \mathbf{e}^T$$

$$\begin{array}{c} \\ \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{array} \begin{array}{c} \begin{array}{ccc} \mathbf{A} & \mathbf{B} & \mathbf{C} \end{array} \\ \begin{bmatrix} 7/15 & 7/15 & 1/3 \\ 7/15 & 1/15 & 1/3 \\ 1/15 & 7/15 & 1/3 \end{bmatrix} \end{array} = \begin{array}{c} \\ \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{array} \begin{array}{c} \begin{array}{ccc} \mathbf{A} & \mathbf{B} & \mathbf{C} \end{array} \\ \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \end{bmatrix} \end{array}$$

Random Teleports (beta = 0.8)



$$\begin{bmatrix} r_A \\ r_B \\ r_C \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \begin{bmatrix} 0.42 \\ 0.29 \\ 0.29 \end{bmatrix} \begin{bmatrix} 0.43 \\ 0.31 \\ 0.26 \end{bmatrix} \begin{bmatrix} 0.43 \\ 0.31 \\ 0.26 \end{bmatrix} \cdots \begin{bmatrix} 35/81 \\ 25/81 \\ 7/27 \end{bmatrix}$$

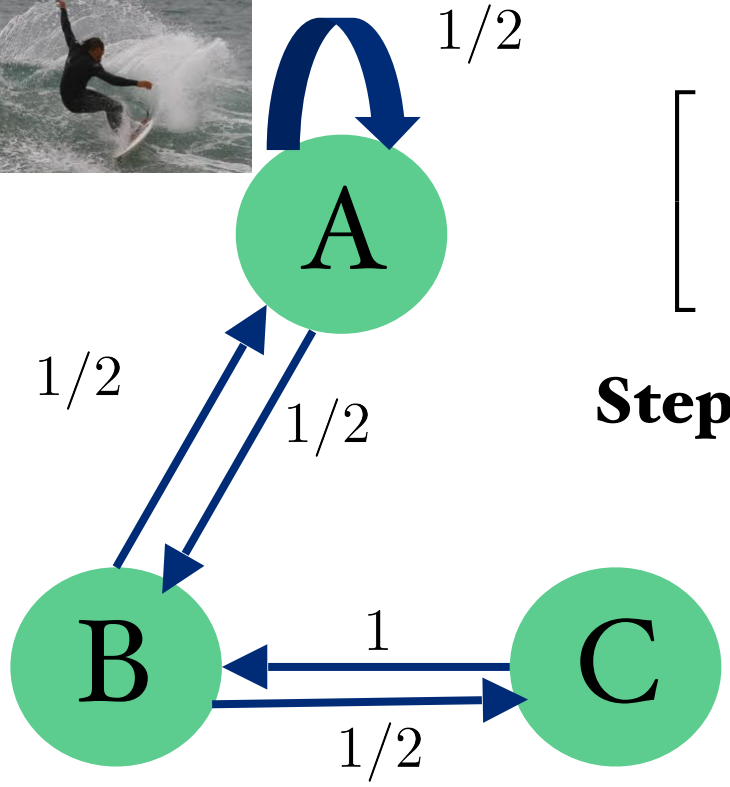
Iteration	$\mathbf{p}$	$\mathbb{G}\mathbf{p}$	$\mathbb{G}^2\mathbf{p}$	$\mathbb{G}^3\mathbf{p}$	...	$\mathbb{G}^\infty\mathbf{p}$
	0	1	2	3	...	*t large*

$$\begin{array}{c} \\ \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{array} \begin{array}{c} \begin{array}{ccc} \mathbf{A} & \mathbf{B} & \mathbf{C} \end{array} \\ \begin{bmatrix} 7/15 & 7/15 & 1/3 \\ 7/15 & 1/15 & 1/3 \\ 1/15 & 7/15 & 1/3 \end{bmatrix} \end{array}$$

Eventually, no website has a zero PageRank score, and the final vector is still a probability vector

Markov Chain Limiting Distribution





$$\begin{array}{c} \mathbf{p} \\ \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \end{array} \begin{array}{c} \mathbb{M}\mathbf{p} \\ \begin{bmatrix} 1/3 \\ 1/2 \\ 1/6 \end{bmatrix} \end{array} \begin{array}{c} \mathbb{M}^2\mathbf{p} \\ \begin{bmatrix} 5/12 \\ 1/3 \\ 1/4 \end{bmatrix} \end{array} \begin{array}{c} \mathbb{M}^3\mathbf{p} \\ \begin{bmatrix} 3/8 \\ 11/24 \\ 1/6 \end{bmatrix} \end{array} \cdots \begin{array}{c} \mathbb{M}^t\mathbf{p} \\ \begin{bmatrix} 37/96 \\ 7/16 \\ 17/96 \end{bmatrix} \end{array} \cdots \begin{array}{c} \mathbb{M}^\infty\mathbf{p} \\ \begin{bmatrix} 2/5 \\ 2/5 \\ 1/5 \end{bmatrix} \end{array} = \boldsymbol{\pi}^*$$

Step	**0**	**1**	**2**	**3**	**...**	***t***	**...**	∞

$$\begin{array}{c} \\ \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{array} \begin{array}{c} \begin{array}{ccc} \mathbf{A} & \mathbf{B} & \mathbf{C} \end{array} \\ \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1 \\ 0 & 1/2 & 0 \end{bmatrix} \end{array} = M$$

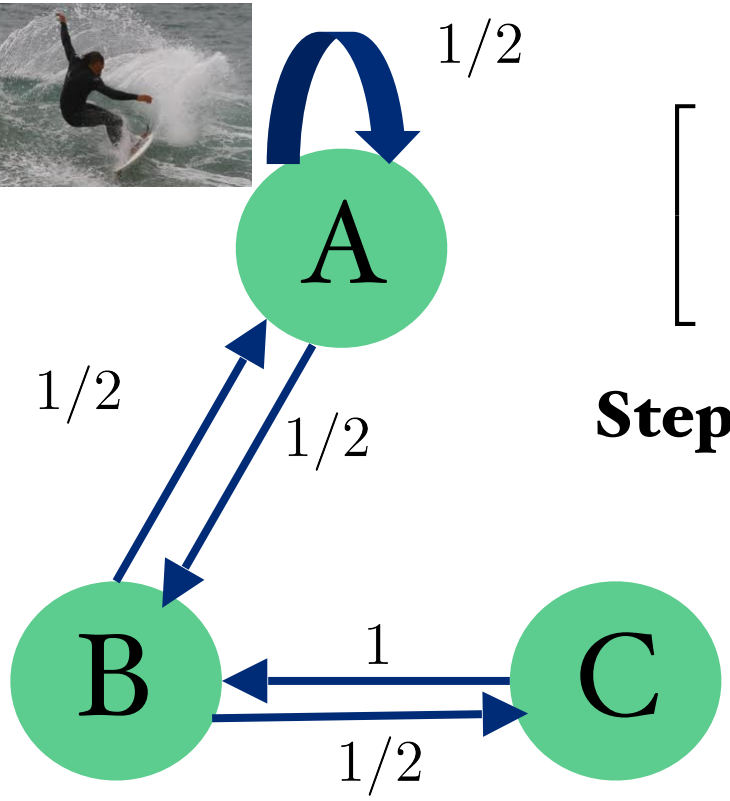
Limiting Distribution:
The limiting distribution is the probability distribution to which the Markov chain converges as the number of steps goes to infinity, if such a distribution exists.

$$\lim_{t \to \infty} \mathbb{M}^t \mathbf{p} = \boldsymbol{\pi}^*$$

where p is the initial distribution (over all states)

Markov Chain Stationary Distribution





$$\begin{array}{c} \mathbf{p} \\ \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \end{array} \begin{array}{c} \mathbb{M}\mathbf{p} \\ \begin{bmatrix} 1/3 \\ 1/2 \\ 1/6 \end{bmatrix} \end{array} \begin{array}{c} \mathbb{M}^2\mathbf{p} \\ \begin{bmatrix} 5/12 \\ 1/3 \\ 1/4 \end{bmatrix} \end{array} \begin{array}{c} \mathbb{M}^3\mathbf{p} \\ \begin{bmatrix} 3/8 \\ 11/24 \\ 1/6 \end{bmatrix} \end{array} \cdots \begin{array}{c} \mathbb{M}^t\mathbf{p} \\ \begin{bmatrix} 37/96 \\ 7/16 \\ 17/96 \end{bmatrix} \end{array} \cdots \begin{array}{c} \mathbb{M}^\infty\mathbf{p} \\ \begin{bmatrix} 2/5 \\ 2/5 \\ 1/5 \end{bmatrix} \end{array} = \boldsymbol{\pi}^*$$

Step	**0**	**1**	**2**	**3**	**...**	***t***	**...**	∞

$$\begin{array}{c} \\ \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{array} \begin{array}{c} \begin{array}{ccc} \mathbf{A} & \mathbf{B} & \mathbf{C} \end{array} \\ \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1 \\ 0 & 1/2 & 0 \end{bmatrix} \end{array} = M$$

Stationary Distribution:

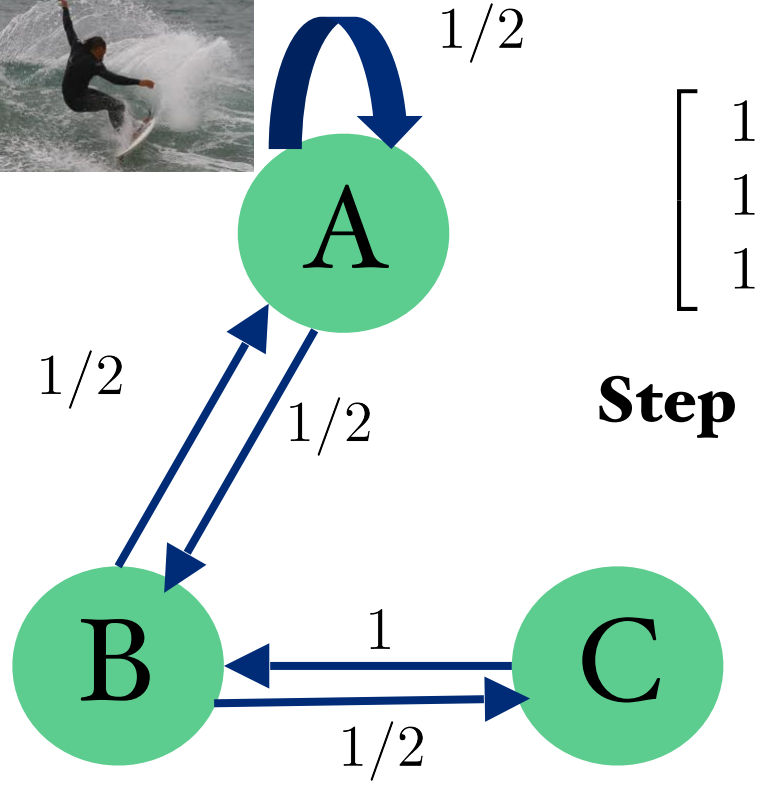
A probability distribution $\boldsymbol{\pi}$ is said to be a stationary distribution for a Markov chain if it satisfies the equation:

$$\boldsymbol{\pi} = \mathbb{M} \cdot \boldsymbol{\pi}$$

If you start the process with the stationary distribution $\boldsymbol{\pi}$, then the probabilities of being in each state remain unchanged after any further steps of transitions

Markov Chain Stationary Distribution vs. Limiting Distribution





$$\mathbf{p} \quad \mathbb{M}\mathbf{p} \quad \mathbb{M}^2\mathbf{p} \quad \mathbb{M}^3\mathbf{p} \quad \mathbb{M}^t\mathbf{p} \quad \mathbb{M}^\infty\mathbf{p}$$

$$\begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \begin{bmatrix} 1/3 \\ 1/2 \\ 1/6 \end{bmatrix} \begin{bmatrix} 5/12 \\ 1/3 \\ 1/4 \end{bmatrix} \begin{bmatrix} 3/8 \\ 11/24 \\ 1/6 \end{bmatrix} \cdots \begin{bmatrix} 37/96 \\ 7/16 \\ 17/96 \end{bmatrix} \cdots \begin{bmatrix} 2/5 \\ 2/5 \\ 1/5 \end{bmatrix} = \boldsymbol{\pi}^*$$

Step 0 1 2 3 ... t ... ∞

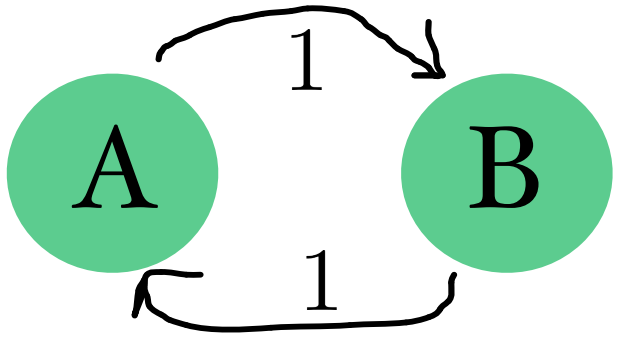
Stationary Distribution

$$\boldsymbol{\pi} = \mathbb{M} \cdot \boldsymbol{\pi}$$

Limiting Distribution

$$\lim_{t \to \infty} \mathbb{M}^t \mathbf{p} = \boldsymbol{\pi}^*$$

$$\begin{array}{c} \\ \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{array} \begin{array}{c} \begin{array}{ccc} \mathbf{A} & \mathbf{B} & \mathbf{C} \end{array} \\ \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1 \\ 0 & 1/2 & 0 \end{bmatrix} \end{array} = M$$



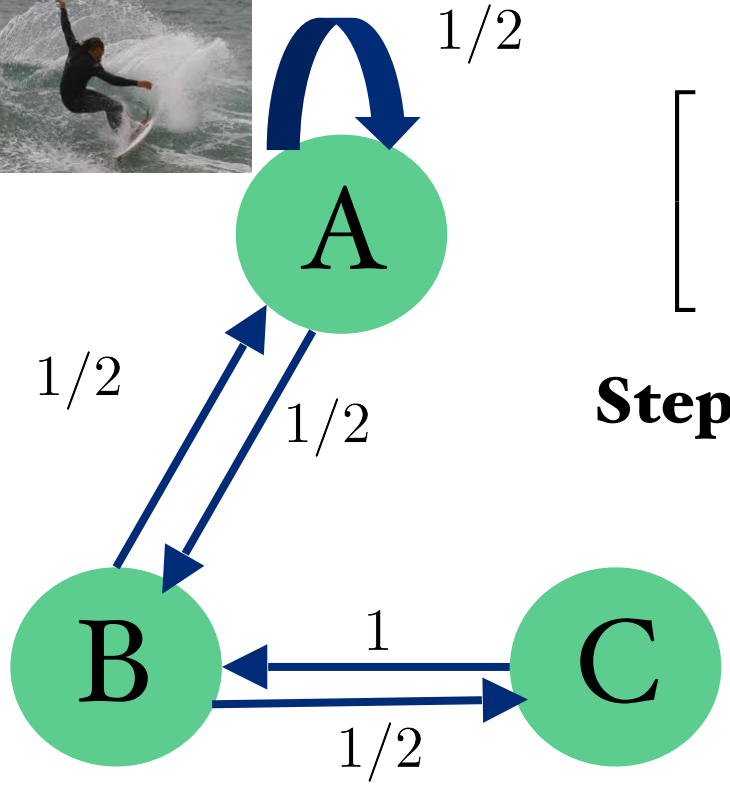
Stationary Distribution

$$\boldsymbol{\pi} = (1/2, 1/2)^T$$

No Limiting Distribution

Markov Chain Long-Run Average Distribution





$$\begin{matrix} \mathbf{p} & \mathbb{M}\mathbf{p} & \mathbb{M}^2\mathbf{p} & \mathbb{M}^3\mathbf{p} & & \mathbb{M}^t\mathbf{p} & & \mathbb{M}^\infty\mathbf{p} \end{matrix}$$
$$\begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \begin{bmatrix} 1/3 \\ 1/2 \\ 1/6 \end{bmatrix} \begin{bmatrix} 5/12 \\ 1/3 \\ 1/4 \end{bmatrix} \begin{bmatrix} 3/8 \\ 11/24 \\ 1/6 \end{bmatrix} \cdots \begin{bmatrix} 37/96 \\ 7/16 \\ 17/96 \end{bmatrix} \cdots \begin{bmatrix} 2/5 \\ 2/5 \\ 1/5 \end{bmatrix} = \boldsymbol{\pi}^*$$

Step	**0**	**1**	**2**	**3**	**...**	***t***	**...**	∞

$$\begin{matrix} & \mathbf{A} & \mathbf{B} & \mathbf{C} \\ \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{matrix} \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1 \\ 0 & 1/2 & 0 \end{bmatrix} = M$$

Long-Run Average Distribution (Long-Run Proportion of Time)

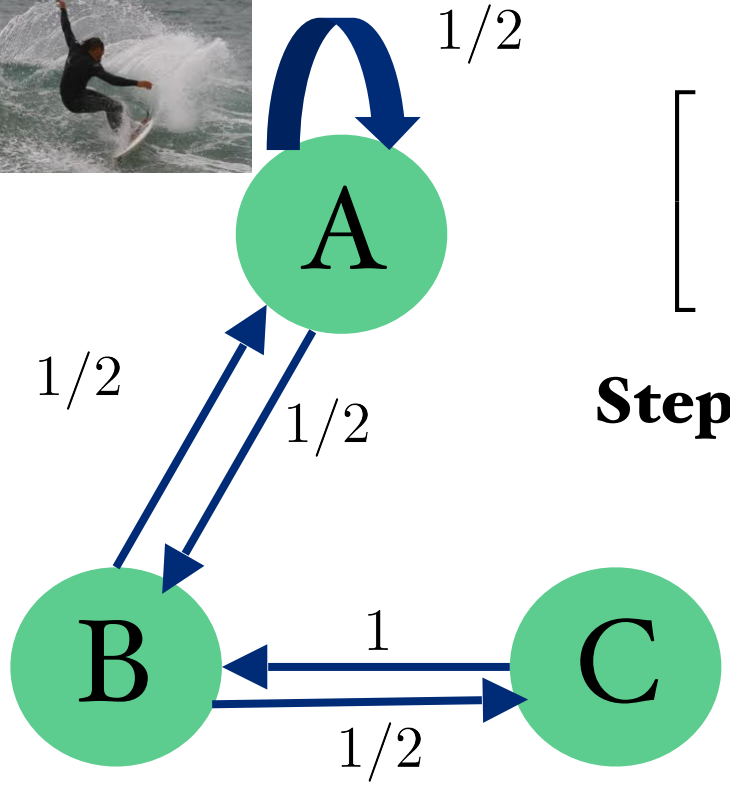
This refers to the fraction of time, as the number of steps goes to infinity, that the Markov chain spends in each state *i*.

$$\lim_{t\to\infty} \frac{1}{t} \sum_{k=0}^{t-1} \mathbb{P}\{X_k = i\}$$

Where X_k is the state of the Markov chain at step *k*.

Markov Chain Stationary Distribution vs. Long-Run Average Distribution





$$\begin{matrix} \mathbf{p} & \mathbb{M}\mathbf{p} & \mathbb{M}^2\mathbf{p} & \mathbb{M}^3\mathbf{p} & & \mathbb{M}^t\mathbf{p} & & \mathbb{M}^\infty\mathbf{p} \end{matrix}$$

$$\begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \begin{bmatrix} 1/3 \\ 1/2 \\ 1/6 \end{bmatrix} \begin{bmatrix} 5/12 \\ 1/3 \\ 1/4 \end{bmatrix} \begin{bmatrix} 3/8 \\ 11/24 \\ 1/6 \end{bmatrix} \cdots \begin{bmatrix} 37/96 \\ 7/16 \\ 17/96 \end{bmatrix} \cdots \begin{bmatrix} 2/5 \\ 2/5 \\ 1/5 \end{bmatrix} = \boldsymbol{\pi}^*$$

Step	**0**	**1**	**2**	**3**	**...**	***t***	**...**	∞

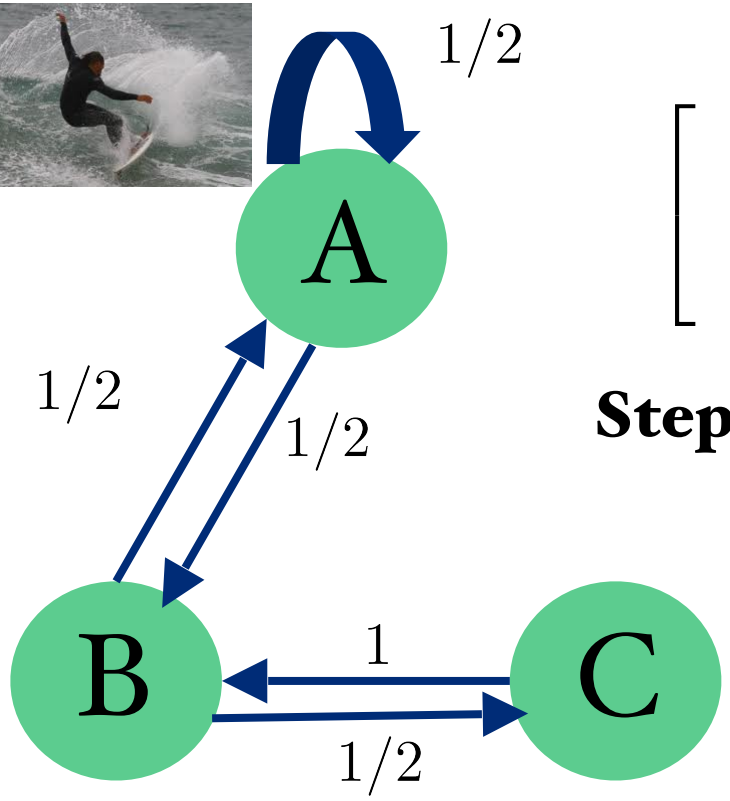
For a Finite-State Markov Chain with all states being accessible from all states

Stationary Distribution $\Longleftrightarrow$ **Long-Run Average Distribution**

$$\begin{matrix} & \mathbf{A} & \mathbf{B} & \mathbf{C} \\ \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{matrix} \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1 \\ 0 & 1/2 & 0 \end{bmatrix} = M$$

Markov Chain Computing Stationary Distribution





$$\begin{matrix} \mathbf{p} & \mathbb{M}\mathbf{p} & \mathbb{M}^2\mathbf{p} & \mathbb{M}^3\mathbf{p} & & \mathbb{M}^t\mathbf{p} & & \mathbb{M}^\infty\mathbf{p} \end{matrix}$$
$$\begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \begin{bmatrix} 1/3 \\ 1/2 \\ 1/6 \end{bmatrix} \begin{bmatrix} 5/12 \\ 1/3 \\ 1/4 \end{bmatrix} \begin{bmatrix} 3/8 \\ 11/24 \\ 1/6 \end{bmatrix} \cdots \begin{bmatrix} 37/96 \\ 7/16 \\ 17/96 \end{bmatrix} \cdots \begin{bmatrix} 2/5 \\ 2/5 \\ 1/5 \end{bmatrix} = \boldsymbol{\pi}^*$$

Step	**0**	**1**	**2**	**3**	**...**	***t***	**...**	∞

$$\begin{matrix} & \mathbf{A} & \mathbf{B} & \mathbf{C} \end{matrix}$$
$$\begin{matrix} \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{matrix} \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1 \\ 0 & 1/2 & 0 \end{bmatrix} = M$$

Computing the Stationary Distribution

Solving $\boldsymbol{\pi} = \mathbb{M} \cdot \boldsymbol{\pi}$ directly ("eigenvalue problem")

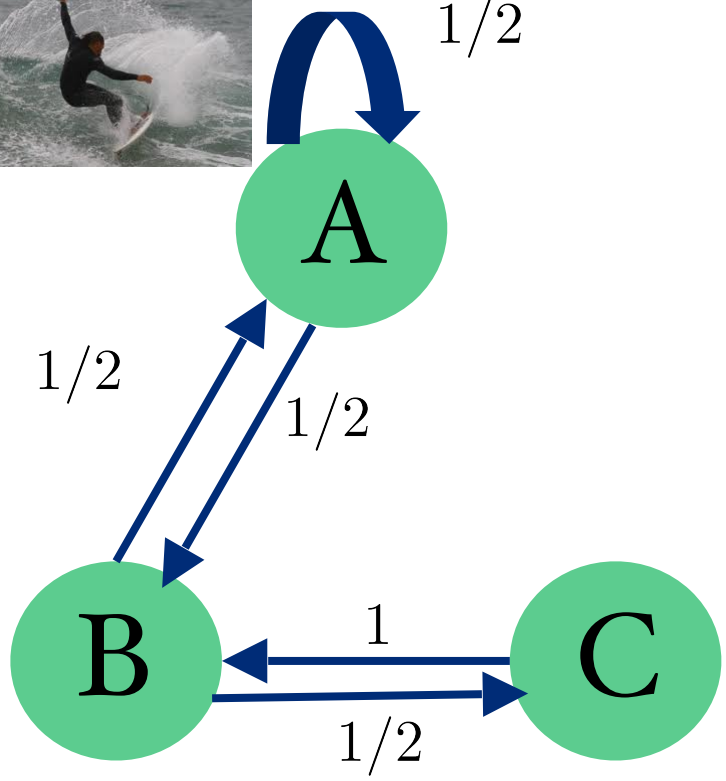
$$s.t. \sum_i \pi = 1, \ \pi_i \geq 0.$$

Compute the Limiting Distribution iteratively

Simulate a long process and estimate the Long-Run Average Distribution

Markov Chain Simulation





$$
\begin{array}{c} \\ \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{array}
\begin{array}{c}
\begin{array}{ccc} \mathbf{A} & \mathbf{B} & \mathbf{C} \end{array} \\
\begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1 \\ 0 & 1/2 & 0 \end{bmatrix}
\end{array}
= M
$$

Set the random seed

Define a vector of size T (let's call it **State**), recording the system state at step $t = 1,2,\ldots, T$

$X =$ 'A'. This is the initial state

For $t = 1$ to T
    State(t) = X;
    R = rand; % a realization of a standard uniform r.v.
    if X=='A'
        if R>1/2, then X= 'B' end
    else
        if X=='B'
            if R>1/2, then X = 'A'; else X = 'C' end
        else
            X = 'B'
        end
end

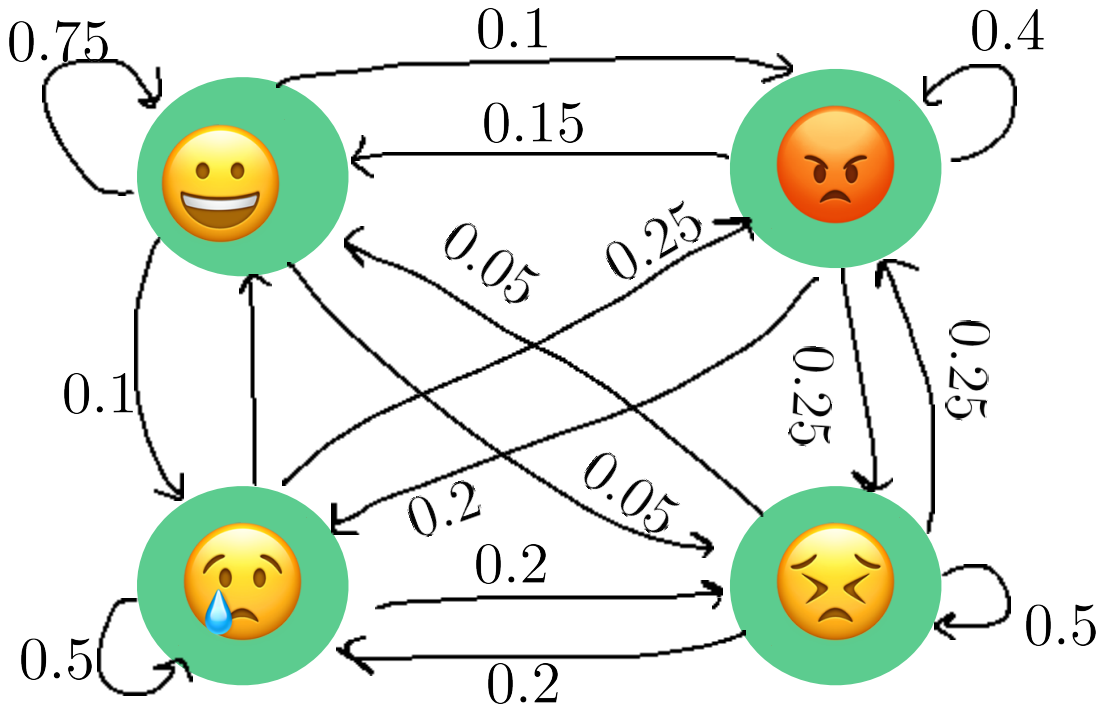
Hidden Markov Model

The "traditional" Markov Model → You can observe all the states

There are settings where system states are unobservable.

What is the most difficult thing to predict/analyze?

	Happy	Sad	Angry	Anxious
Happy	0.75	0.1	0.1	0.05
Sad	0.05	0.5	0.25	0.2
Angry	0.15	0.2	0.4	0.25
Anxious	0.05	0.2	0.25	0.5



Hidden Markov Model

Your significant other's shopping pattern…

In this example, mood is the "hidden state"

Illustrative emission probabilities: Pr{shopping code|mood}

	($0,$1000), include non-essentials (S1)	($0,$1000), include only essentials (S2)	>=$1000 (S3)	$0 (S4)
Happy	**80%**	**10%**	**5%**	**5%**
Sad	**10%**	**10%**	**5%**	**75%**
Angry	**5%**	**15%**	**65%**	**15%**
Anxious	**5%**	**65%**	**25%**	**5%**

Hidden Markov Model

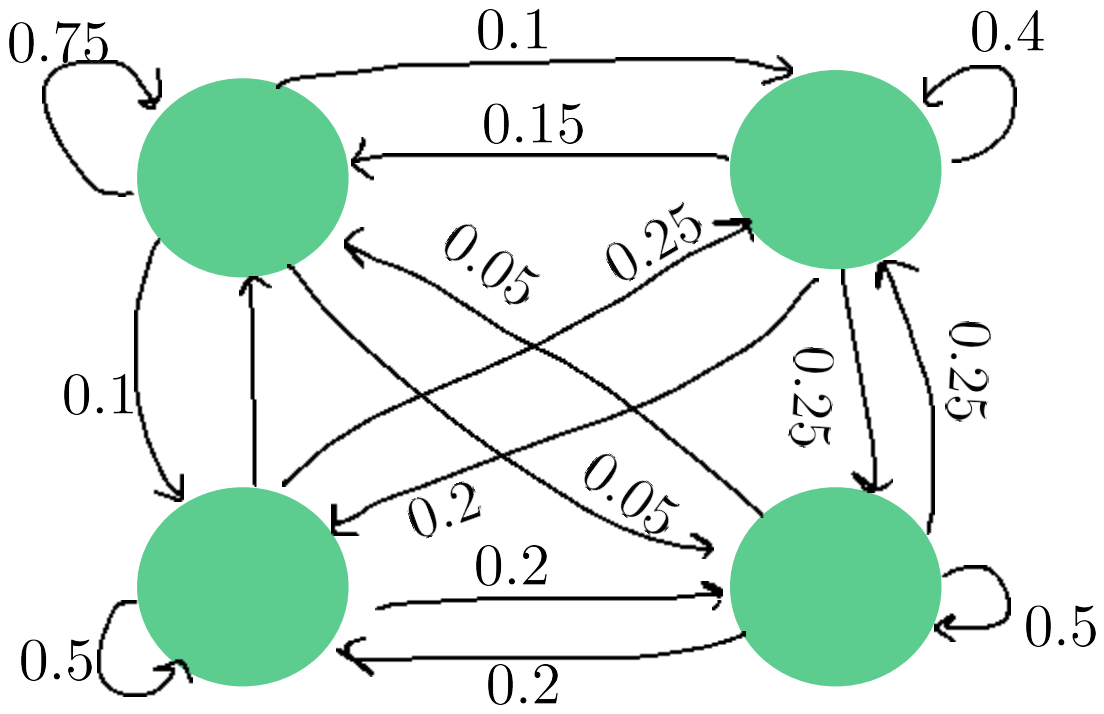
Your significant other's shopping pattern...

In this example, mood is the "hidden state"

With this model, I can predict the mood sequence if I know the shopping pattern.

Viterbi Search Algorithm

$$\mathbb{P}\{S_1|Happy\} = 0.8$$
$$\mathbb{P}\{S_2|Happy\} = 0.1$$
$$\mathbb{P}\{S_3|Happy\} = 0.05$$
$$\mathbb{P}\{S_4|Happy\} = 0.05$$



$$\mathbb{P}\{S_1|Angry\} = 0.05$$
$$\mathbb{P}\{S_2|Angry\} = 0.15$$
$$\mathbb{P}\{S_3|Angry\} = 0.65$$
$$\mathbb{P}\{S_4|Angry\} = 0.15$$

$$\mathbb{P}\{S_1|Sad\} = 0.1$$
$$\mathbb{P}\{S_2|Sad\} = 0.1$$
$$\mathbb{P}\{S_3|Sad\} = 0.05$$
$$\mathbb{P}\{S_4|Sad\} = 0.75$$

$$\mathbb{P}\{S_1|Anxious\} = 0.05$$
$$\mathbb{P}\{S_2|Anxious\} = 0.65$$
$$\mathbb{P}\{S_3|Anxious\} = 0.25$$
$$\mathbb{P}\{S_4|Anxious\} = 0.05$$